Roll No……..

MID SEMESTER EXAMINATION MARCH 2025

Name of the Program: B.Tech(ECE) *Semester: IV*

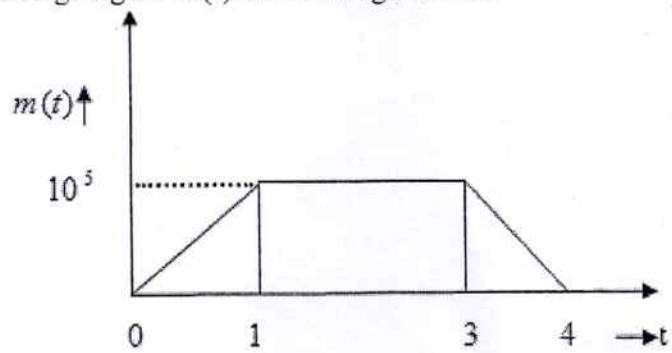
Name of the Course: Communication Systems I *Course Code:* TEC 401

Time: 90 Minutes Maximum Marks: 50

Note:

(i) Answer **all the questions** by choosing ***any one of the sub questions***.

(ii) Each question carries 10 marks

Q1	(10 marks)	**CO1**
(a)	Obtain an expression for modulation index of an AM wave using oscilloscope method.	
	OR	
(b)	(i) Calculate the percentage power saving when the carrier and one of the sidebands are suppressed in an AM wave modulated to a depth of (a) 100 percent and (b) 50 percent. (ii) Show that a signal and its Hilbert transform are mutually orthogonal.	
Q2	(10 marks)	**CO1**
(a)	Draw the block diagram of communication system and explain the function of each block.	
	OR	
(b)	Draw the circuit diagram of a square law amplitude modulator and explain its working.	
Q3	(10 marks)	**CO1**
(a)	An amplitude modulated signal is given by $s(t) = 10\cos(2\pi 10^6 t) + 5\cos(2\pi 10^6 t)\times\cos(2\pi 10^3 t) + 2\cos(2\pi 10^6 t)\times\cos(4\pi 10^3 t)$ volt. Find the net modulation index and total transmitted power.	
	OR	
(b)	Explain demodulation of DSB-SC wave using synchronous detector. Also, discuss the frequency and phase errors in synchronous detector.	
Q4	(10 marks)	**CO1/ CO2**
(a)	What is diagonal clipping in amplitude modulation and how does the RC time constant decide this phenomenon?	
	OR	
(b)	Derive an expression for Carson's rule approximation in FM wave.	
Q5	(10 marks)	**CO2**
(a)	For the message signal $m(t)$ shown in figure below  	

	(i) If $m(t)$ is frequency modulated on a carrier with frequency 10^6 Hz, with a frequency deviation constant equal to 5 Hz/volt, what is the maximum instantaneous frequency of the modulated signal. (ii) If $m(t)$ is phase modulated on a carrier with frequency 10^6 Hz, with a phase deviation constant equal to 3 rad/volt, what is the maximum instantaneous frequency of the modulated signal.	
	OR	
(b)	(i) A signal $x(t) = 2\cos(2\pi \times 52\, t)$ volts is applied to an FM modulator with the frequency sensitivity factor of 20 KHz/volt. Determine the modulation index of the FM wave. (ii) An angle modulated signal is described by $S(t) = 5\cos[2\pi 10^6 t + 200\sin(10^3 \pi t)]$. Considering $S(t)$ as a PM signal with $k_p = 100$ rad/volt, find the modulating signal $m(t)$.	